

Neuroscience and AI in Sensation and Motivation

In Partnership With University of Southern California

April 23–24, 2026 Zilkha Neurogenetic Institute of Keck School of Medicine of USC, Los Angeles
Organized by Dr. Li Zhang, Center for Neural Circuits & Sensory Processing Disorders

ABOUT THE SYMPOSIA

Building on the ongoing partnership between the Chen Institute and USC, the 4th Annual Symposium on Sensation, Motivation, and AI will focus on how brain circuits integrate external and internal sensory information to drive motivated behaviors. The symposium will also highlight the growing interface between neuroscience and artificial intelligence, emphasizing the integration of AI concepts and methodologies with neuroscience research on brain function and behavior. By bringing together researchers from diverse disciplines, the event aims to explore how brain circuits produce sensation and motivation while fostering new interdisciplinary collaborations.

April 23, 2026 9:00AM - 5:00PM

9:00AM Check-In and Breakfast

Session 1: Sensation, Motivation, and Neural Circuits

9:40AM Michael Halassa Thalamocortical Interactions In Cognitive Control And Flexibility
10:10AM Laura Busse Thalamo-Cortico-Thalamic Loops: Shaping Visual Processing And Behavior
10:40AM Brice Bathellier A Deep Encoder For Sensory Restoration By Neuromimetic Cortical Stimulation
11:10AM Coffee Break
11:20AM Lauren McElvain Re-evaluating Basal Ganglia Connectivity: From Anatomy to Action
11:50AM Herbert Wu Neural Basis Of Leader-Follower Dynamics In Cooperative Behavior
12:20PM Guang-Wei Zhang Understanding Animal Behavior From Video
12:50PM Lunch

Session 2: Neuroscience and Artificial Intelligence

2:10PM Bing Wen Brunton Embodied Intelligence: From Connectomes To Body Models
2:40PM Brian Lee Directional Hand Movement Can Be Classified From Insular Cortex SEEG Signals Using Recurrent Neural Networks And High-Gamma Band Features
3:10PM Daniel L. K. Yamins World Modeling With Probabilistic Structure Integration
3:40PM Coffee Break
3:50PM Samuel Nastase Unifying The Structures Of Language In A Neural Population Code
4:20PM Talmo Pereira Towards A Science Of Neuroconnectionist Ethology
4:50PM Reception

April 24, 2026 9:00AM - 5:00PM

Private Speakers Workshop - Invitation Only

Program



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